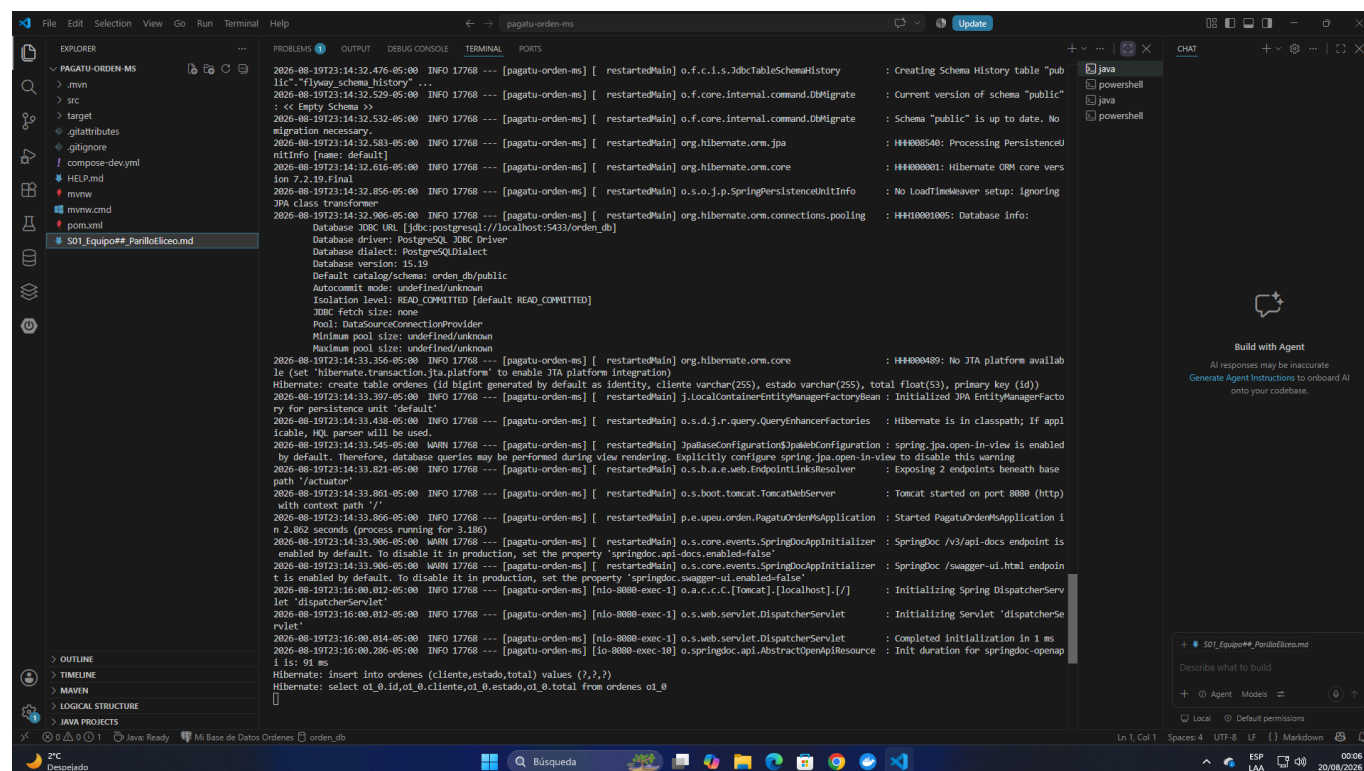


Actividad Autónoma: Replicación de Microservicio (orden-ms)

Estudiante: Eliceo Parillo Mostajo

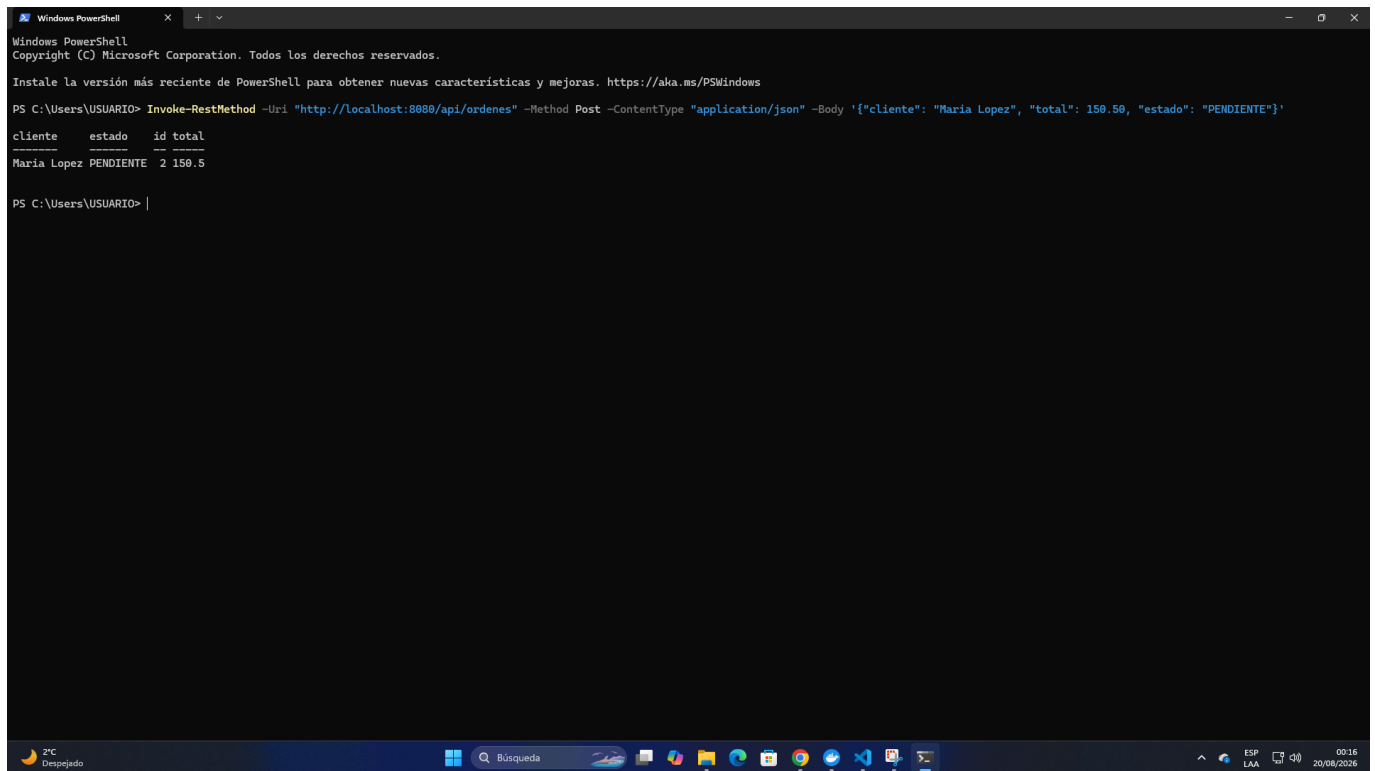
Equipo: ##

1. Ejecutar el microservicio en DEV con Maven Wrapper



```
2026-08-19T23:14:32.476-05:00 INFO 17768 --- [pagatu-orden-ms] [ restartedMain] o.f.c.i.s.JdbcTableSchemaHistory : Creating Schema History table "public"
2026-08-19T23:14:32.529-05:00 INFO 17768 --- [pagatu-orden-ms] [ restartedMain] o.f.core.internal.command.DBMigrate : Current version of schema "public"
2026-08-19T23:14:32.532-05:00 INFO 17768 --- [pagatu-orden-ms] [ restartedMain] o.f.core.internal.command.DBMigrate : Schema "public" is up to date. No
2026-08-19T23:14:32.583-05:00 INFO 17768 --- [pagatu-orden-ms] [ restartedMain] org.hibernate.orm.jpa : HHH000540: Processing PersistenceU
2026-08-19T23:14:32.616-05:00 INFO 17768 --- [pagatu-orden-ms] [ restartedMain] org.hibernate.orm.core : HHH000001: Hibernate ORM core vers
2026-08-19T23:14:32.966-05:00 INFO 17768 --- [pagatu-orden-ms] [ restartedMain] org.hibernate.orm.connections.pooling : HHH10001005: Database info:
Database JDBC URL [jdbc:postgresql://localhost:5433/orden_db]
Database driver PostgreSQL JDBC Driver
Database dialect: PostgreSQLDialect
Database version: 15.19
Default catalog/schema: orden_db/public
Automatic mode: undefined/unknown
Isolation level: READ COMMITTED [default READ_COMMITTED]
JDBC fetch size: none
Pool: DataSourceConnectionProvider
Minimum pool size: undefined/unknown
Maximum pool size: undefined/unknown
2026-08-19T23:14:33.356-05:00 INFO 17768 --- [pagatu-orden-ms] [ restartedMain] org.hibernate.orm.core : HHH000489: No JTA platform avail
le (set "hibernate.transaction.jta.platform" to enable JTA platform integration)
Hibernate: create table ordenes (id bigint generated by default as identity, cliente varchar(255), estado varchar(255), total float(53), primary key (id))
2026-08-19T23:14:33.397-05:00 INFO 17768 --- [pagatu-orden-ms] [ restartedMain] j.LocalContainerEntityManagerFactoryBean : Initialized JPA EntityManagerFacto
ry for persistence unit 'default'
2026-08-19T23:14:33.438-05:00 INFO 17768 --- [pagatu-orden-ms] [ restartedMain] o.s.d.j.r.query.QueryEnhancerFactories : Hibernate is in classpath; If appl
icable, SQL parser will be used.
2026-08-19T23:14:33.545-05:00 WARN 17768 --- [pagatu-orden-ms] [ restartedMain] jpabaseConfiguration$JpakebConfiguration : spring.jpa.open-in-view is enabled
by default. Therefore, database queries may be performed during view rendering. Explicitly configure spring.jpa.open-in-view to disable this warning
2026-08-19T23:14:33.621-05:00 INFO 17768 --- [pagatu-orden-ms] [ restartedMain] o.s.b.a.e.web.EndpointLinksResolver : Exposing 2 endpoints beneath base
path '/actuator'
2026-08-19T23:14:33.861-05:00 INFO 17768 --- [pagatu-orden-ms] [ restartedMain] o.s.boot.tomcat.TomcatWebServer : Tomcat started on port 8080 (http)
with context path '/'
2026-08-19T23:14:33.966-05:00 INFO 17768 --- [pagatu-orden-ms] [ restartedMain] p.e.uepu.orden.PagatuOrderMsApplication : Started PagatuOrderMsApplication i
n 2.882 seconds (process running for 3.186)
2026-08-19T23:14:33.969-05:00 WARN 17768 --- [pagatu-orden-ms] [ restartedMain] o.s.core.events.SpringDocAppInitializer : SpringDoc /v3/api-docs endpoint is
enabled by default. To disable it in production, set the property 'springdoc.api-docs.enabled=false'
2026-08-19T23:14:33.986-05:00 WARN 17768 --- [pagatu-orden-ms] [ restartedMain] o.s.core.events.SpringDocAppInitializer : SpringDoc /swagger-ui.html endpoi
nt is enabled by default. To disable it in production, set the property 'springdoc.swagger-ui.enabled=false'
2026-08-19T23:16:00.012-05:00 INFO 17768 --- [pagatu-orden-ms] [nio-8080-exec-1] o.a.c.c.C.[Tomcat].[localhost].[/] : Initializing Spring DispatcherServ
let 'dispatcherServlet'
2026-08-19T23:16:00.012-05:00 INFO 17768 --- [pagatu-orden-ms] [nio-8080-exec-1] o.s.web.servlet.DispatcherServlet : Initializing Servlet 'dispatcherSe
rvlet'
2026-08-19T23:16:00.014-05:00 INFO 17768 --- [pagatu-orden-ms] [nio-8080-exec-1] o.s.web.servlet.DispatcherServlet : Completed initialization in 1 ms
2026-08-19T23:16:00.286-05:00 INFO 17768 --- [pagatu-orden-ms] [io-8080-exec-10] o.springdoc.api.AbstractOpenApiResource : Init duration for springdoc-openap
i is: 94 ms
Hibernate: Insert into ordenes (cliente,estado,total) values (?,?,?)
Hibernate: select oi_0.id,oi_0.cliente,oi_0.estado,oi_0.total from ordenes oi_0
```

2. Probar el CRUD por PowerShell



```
Windows PowerShell
Copyright (C) Microsoft Corporation. Todos los derechos reservados.

Instale la versión más reciente de PowerShell para obtener nuevas características y mejoras. https://aka.ms/PSWindows

PS C:\Users\USUARIO> Invoke-RestMethod -Uri "http://localhost:8088/api/ordenes" -Method Post -ContentType "application/json" -Body '{"cliente": "Maria Lopez", "total": 150.50, "estado": "PENDIENTE"}'

cliente      estado      id total
-----
Maria Lopez  PENDIENTE   2 150.5

PS C:\Users\USUARIO> |
```

The screenshot shows a Windows PowerShell terminal window. The title bar reads "Windows PowerShell". The command prompt shows the user running an `Invoke-RestMethod` command to send a POST request to `http://localhost:8088/api/ordenes` with a JSON body. The output is a table with columns `cliente`, `estado`, `id`, and `total`. The data row shows `Maria Lopez`, `PENDIENTE`, `2`, and `150.5`. The taskbar at the bottom shows the system clock as 00:16 on 20/08/2026.

3. Verificar Swagger, /actuator/health y /actuator/metrics

The first screenshot shows the Swagger UI for the 'orden-controller' API. It displays the 'POST /api/ordenes' endpoint. The 'Parameters' section indicates 'No parameters'. The 'Request body' section is marked as 'required' and shows an 'application/json' media type. An 'Example Value' is provided:

```
{  "id": 0,  "cliente": "string",  "total": 0.1,  "estado": "string"}
```

. The 'Responses' section shows a '200 OK' status with a 'Media type' of 'application/json' and an 'Example Value' of

```
{  "id": 0,  "cliente": "string",  "total": 0.1,  "estado": "string"}
```

.

The second screenshot shows the Swagger UI for the 'localhost:8080/actuator' endpoint. It displays the 'GET /actuator' endpoint. The 'Parameters' section indicates 'No parameters'. The 'Request body' section is marked as 'required' and shows an 'application/json' media type. An 'Example Value' is provided:

```
{  "id": 0,  "cliente": "string",  "total": 0.1,  "estado": "string"}
```

. The 'Responses' section shows a '200 OK' status with a 'Media type' of 'application/json' and an 'Example Value' of

```
{  "id": 0,  "cliente": "string",  "total": 0.1,  "estado": "string"}
```

.

4. Revisar la base de datos

The screenshot shows the Visual Studio Code interface with a database extension. The left sidebar displays the 'DATABASE' explorer with a tree view showing 'Mi Base de Datos Ordenes' and its sub-items: 'Security', 'orden_db', 'Query', 'public 56K', 'Query', 'Tables (2)', 'flyway_schema_history', and 'ordenes'. The main editor area shows a SQL query: `SELECT * FROM ordenes LIMIT 100`. Below the query, the results are displayed in a table with columns: 'id', 'cliente', 'estado', and 'total'. The results show two rows:
1 | Maria Lopez | PAGADO | 250
2 | Maria Lopez | PENDIENTE | 150.5
The bottom panel shows the 'TERMINAL' with logs from the application, including messages about schema migration, Hibernate ORM core version, and database info. The right sidebar shows a 'CHAT' panel with a 'Build with Agent' button and a 'S01_Equipo##_ParilloEliceo.md' file.

id	cliente	estado	total
1	Maria Lopez	PAGADO	250
2	Maria Lopez	PENDIENTE	150.5

5. Ejecutar dos instancias en paralelo (8080 y 8081)

```

2026-08-19T23:40:54.586-05:00 INFO 22272 --- [pagatu-orden-ms] [ restartedMain] org.flywaydb.core.FlywayExecutor : Database: jdbc:postgresql://localhost:5433/orden_db (PostgreSQL 15.19)
2026-08-19T23:40:54.620-05:00 INFO 22272 --- [pagatu-orden-ms] [ restartedMain] o.f.core.internal.command.DbValidate : Successfully validated 0 migrations (execution time 00:00:01.15)
2026-08-19T23:40:54.620-05:00 WARN 22272 --- [pagatu-orden-ms] [ restartedMain] o.f.core.internal.command.DbValidate : No migrations found. Are your locations set up correctly?
2026-08-19T23:40:54.645-05:00 INFO 22272 --- [pagatu-orden-ms] [ restartedMain] o.f.core.internal.command.DbMigrate : Current version of schema "public" is << Empty Schema >>
2026-08-19T23:40:54.648-05:00 INFO 22272 --- [pagatu-orden-ms] [ restartedMain] o.f.core.internal.command.DbMigrate : Schema "public" is up to date. No migration necessary.
2026-08-19T23:40:54.704-05:00 INFO 22272 --- [pagatu-orden-ms] [ restartedMain] org.hibernate.orm.jpa : HHH000540: Processing PersistenceUnitInfo [name: default]
2026-08-19T23:40:54.733-05:00 INFO 22272 --- [pagatu-orden-ms] [ restartedMain] org.hibernate.orm.core : HHH000001: Hibernate ORM core version 7.2.19.Final
2026-08-19T23:40:54.968-05:00 INFO 22272 --- [pagatu-orden-ms] [ restartedMain] o.s.o.j.p.SpringPersistenceUnitInfo : No LoadTimeWeaver setup; ignoring
2026-08-19T23:40:55.019-05:00 INFO 22272 --- [pagatu-orden-ms] [ restartedMain] org.hibernate.orm.connections.pooling : HHH10001005: Database info:
Database JDBC URL [jdbc:postgresql://localhost:5433/orden_db]
Database driver: PostgreSQL JDBC Driver
Database dialect: PostgreSQLDialect
Database version: 15.19
Default catalog/schema: orden_db/public
Autocommit mode: undefined/unknown
Isolation level: READ COMMITTED [default READ COMMITTED]
JDBC fetch size: none
Pool: DataSourceConnectionProvider
Minimum pool size: undefined/unknown
Maximum pool size: undefined/unknown
2026-08-19T23:40:55.472-05:00 INFO 22272 --- [pagatu-orden-ms] [ restartedMain] org.hibernate.orm.core : HHH000489: No JTA platform available (set 'hibernate.transaction.jta.platform' to enable JTA platform integration)
2026-08-19T23:40:55.581-05:00 INFO 22272 --- [pagatu-orden-ms] [ restartedMain] j.LocalContainerEntityManagerFactoryBean : Initialized JPA EntityManagerFactory for persistence unit 'default'
2026-08-19T23:40:55.545-05:00 INFO 22272 --- [pagatu-orden-ms] [ restartedMain] o.s.d.j.r.query.QueryEnhancerFactories : Hibernate is in classpath; If applicable, HQL parser will be used.
2026-08-19T23:40:55.638-05:00 WARN 22272 --- [pagatu-orden-ms] [ restartedMain] jpaBaseConfiguration$JpaWebConfiguration : spring.jpa.open-in-view is enabled by default. Therefore, database queries may be performed during view rendering. Explicitly configure spring.jpa.open-in-view to disable this warning
2026-08-19T23:40:55.908-05:00 INFO 22272 --- [pagatu-orden-ms] [ restartedMain] o.s.b.a.e.web.EndpointLinksResolver : Exposing 2 endpoints beneath base path '/actuator'
2026-08-19T23:40:55.973-05:00 INFO 22272 --- [pagatu-orden-ms] [ restartedMain] o.s.boot.tomcat.TomcatWebServer : Tomcat started on port 8081 (http) with context path '/'
2026-08-19T23:40:55.978-05:00 INFO 22272 --- [pagatu-orden-ms] [ restartedMain] p.e.upsu.orden.PagatuOrdenMsApplication : Started PagatuOrdenMsApplication in 2.816 seconds (process running for 3.887)
2026-08-19T23:40:56.014-05:00 WARN 22272 --- [pagatu-orden-ms] [ restartedMain] o.s.core.events.SpringDocAppInitializer : SpringDoc /v3/api-docs endpoint is enabled by default. To disable it in production, set the property 'springdoc.api-docs.enabled=false'
2026-08-19T23:40:56.014-05:00 WARN 22272 --- [pagatu-orden-ms] [ restartedMain] o.s.core.events.SpringDocAppInitializer : SpringDoc /swagger-ui.html endpoint is enabled by default. To disable it in production, set the property 'springdoc.swagger-ui.enabled=false'
2026-08-19T23:41:57.157-05:00 INFO 22272 --- [pagatu-orden-ms] [nio-8081-exec-2] o.a.c.c.c.[tomcat].[localhost].[/] : Initializing Spring DispatcherServlet 'dispatcherServlet'
2026-08-19T23:41:57.157-05:00 INFO 22272 --- [pagatu-orden-ms] [nio-8081-exec-2] o.s.web.servlet.DispatcherServlet : Initializing Servlet 'dispatcherServlet'
2026-08-19T23:41:57.158-05:00 INFO 22272 --- [pagatu-orden-ms] [nio-8081-exec-2] o.s.web.servlet.DispatcherServlet : Completed initialization in 1 ms
2026-08-19T23:41:57.449-05:00 INFO 22272 --- [pagatu-orden-ms] [nio-8081-exec-10] o.springdoc.api.AbstractOpenApiResource : Init duration for springdoc-openapi is: 95 ms

```

```

2026-08-19T23:14:32.476-05:00 INFO 17768 --- [pagatu-orden-ms] [ restartedMain] o.f.c.i.s.JdbcTableSchemaHistory : Creating Schema History table "public"."flyway_schema_history"
2026-08-19T23:14:32.529-05:00 INFO 17768 --- [pagatu-orden-ms] [ restartedMain] o.f.core.internal.command.DbMigrate : Current version of schema "public" is << Empty Schema >>
2026-08-19T23:14:32.532-05:00 INFO 17768 --- [pagatu-orden-ms] [ restartedMain] o.f.core.internal.command.DbMigrate : Schema "public" is up to date. No migration necessary.
2026-08-19T23:14:32.583-05:00 INFO 17768 --- [pagatu-orden-ms] [ restartedMain] org.hibernate.orm.jpa : HHH000540: Processing PersistenceUnitInfo [name: default]
2026-08-19T23:14:32.616-05:00 INFO 17768 --- [pagatu-orden-ms] [ restartedMain] org.hibernate.orm.core : HHH000001: Hibernate ORM core version 7.2.19.Final
2026-08-19T23:14:32.856-05:00 INFO 17768 --- [pagatu-orden-ms] [ restartedMain] o.s.o.j.p.SpringPersistenceUnitInfo : No LoadTimeWeaver setup; ignoring
2026-08-19T23:14:32.906-05:00 INFO 17768 --- [pagatu-orden-ms] [ restartedMain] org.hibernate.orm.connections.pooling : HHH10001005: Database info:
Database JDBC URL [jdbc:postgresql://localhost:5433/orden_db]
Database driver: PostgreSQL JDBC Driver
Database dialect: PostgreSQLDialect
Database version: 15.19
Default catalog/schema: orden_db/public
Autocommit mode: undefined/unknown
Isolation level: READ COMMITTED [default READ COMMITTED]
JDBC fetch size: none
Pool: DataSourceConnectionProvider
Minimum pool size: undefined/unknown
Maximum pool size: undefined/unknown
2026-08-19T23:14:33.356-05:00 INFO 17768 --- [pagatu-orden-ms] [ restartedMain] org.hibernate.orm.core : HHH000489: No JTA platform available (set 'hibernate.transaction.jta.platform' to enable JTA platform integration)
2026-08-19T23:14:33.397-05:00 INFO 17768 --- [pagatu-orden-ms] [ restartedMain] j.LocalContainerEntityManagerFactoryBean : Initialized JPA EntityManagerFactory for persistence unit 'default'
2026-08-19T23:14:33.438-05:00 INFO 17768 --- [pagatu-orden-ms] [ restartedMain] o.s.d.j.r.query.QueryEnhancerFactories : Hibernate is in classpath; If applicable, HQL parser will be used.
2026-08-19T23:14:33.545-05:00 WARN 17768 --- [pagatu-orden-ms] [ restartedMain] jpaBaseConfiguration$JpaWebConfiguration : spring.jpa.open-in-view is enabled by default. Therefore, database queries may be performed during view rendering. Explicitly configure spring.jpa.open-in-view to disable this warning
2026-08-19T23:14:33.821-05:00 INFO 17768 --- [pagatu-orden-ms] [ restartedMain] o.s.b.a.e.web.EndpointLinksResolver : Exposing 2 endpoints beneath base path '/actuator'
2026-08-19T23:14:33.861-05:00 INFO 17768 --- [pagatu-orden-ms] [ restartedMain] o.s.boot.tomcat.TomcatWebServer : Tomcat started on port 8080 (http) with context path '/'
2026-08-19T23:14:33.865-05:00 INFO 17768 --- [pagatu-orden-ms] [ restartedMain] p.e.upsu.orden.PagatuOrdenMsApplication : Started PagatuOrdenMsApplication in 2.862 seconds (process running for 3.186)
2026-08-19T23:14:33.906-05:00 WARN 17768 --- [pagatu-orden-ms] [ restartedMain] o.s.core.events.SpringDocAppInitializer : SpringDoc /v3/api-docs endpoint is enabled by default. To disable it in production, set the property 'springdoc.api-docs.enabled=false'
2026-08-19T23:14:33.906-05:00 WARN 17768 --- [pagatu-orden-ms] [ restartedMain] o.s.core.events.SpringDocAppInitializer : SpringDoc /swagger-ui.html endpoint is enabled by default. To disable it in production, set the property 'springdoc.swagger-ui.enabled=false'
2026-08-19T23:16:00.012-05:00 INFO 17768 --- [pagatu-orden-ms] [nio-8080-exec-1] o.a.c.c.c.[tomcat].[localhost].[/] : Initializing Spring DispatcherServlet 'dispatcherServlet'
2026-08-19T23:16:00.012-05:00 INFO 17768 --- [pagatu-orden-ms] [nio-8080-exec-1] o.s.web.servlet.DispatcherServlet : Initializing Servlet 'dispatcherServlet'
2026-08-19T23:16:00.014-05:00 INFO 17768 --- [pagatu-orden-ms] [nio-8080-exec-1] o.s.web.servlet.DispatcherServlet : Completed initialization in 1 ms
2026-08-19T23:16:00.286-05:00 INFO 17768 --- [pagatu-orden-ms] [nio-8080-exec-10] o.springdoc.api.AbstractOpenApiResource : Init duration for springdoc-openapi is: 91 ms
Hibernate: insert into ordenes (cliente,estado,total) values (?,?,?)
Hibernate: select o_0.id,o_0.cliente,o_0.estado,o_0.total from ordenes o_0
Hibernate: insert into ordenes (cliente,estado,total) values (?,?,?)

```

6. ¿Por qué un microservicio debe poder escalar horizontalmente sin puerto fijo?

El escalado horizontal dinámico es un pilar fundamental en la arquitectura de microservicios por las siguientes razones técnicas:

- **Prevención de conflictos de puertos:** En entornos de producción (contenedores Docker/Kubernetes), múltiples instancias residen en el mismo host. Asignar un puerto estático causaría conflictos de vinculación, impidiendo el despliegue automático.
- **Abstracción mediante API Gateway:** La escalabilidad requiere que el cliente ignore la dirección IP o puerto específico de una instancia. El uso de puertos dinámicos obliga a integrar un **API Gateway** o

balanceador de carga, que actúa como punto de entrada único, enrutando el tráfico hacia las instancias saludables.

- **Elasticidad y Agilidad:** Desplegar instancias en puertos arbitrarios permite que el orquestador gestione el ciclo de vida del servicio (creación, destrucción, recuperación ante fallos) de forma transparente, optimizando la disponibilidad y la eficiencia de los recursos.